

ETAS-Inspired Spatio-Temporal Convolutional (STC) Model for Next-Day Earthquake Forecasting

Chengxiang Zhan, Shichen Gao, Ying Zhang, Jiawei Li, and Qingyan Meng

zhancx@email.cugb.edu.cn

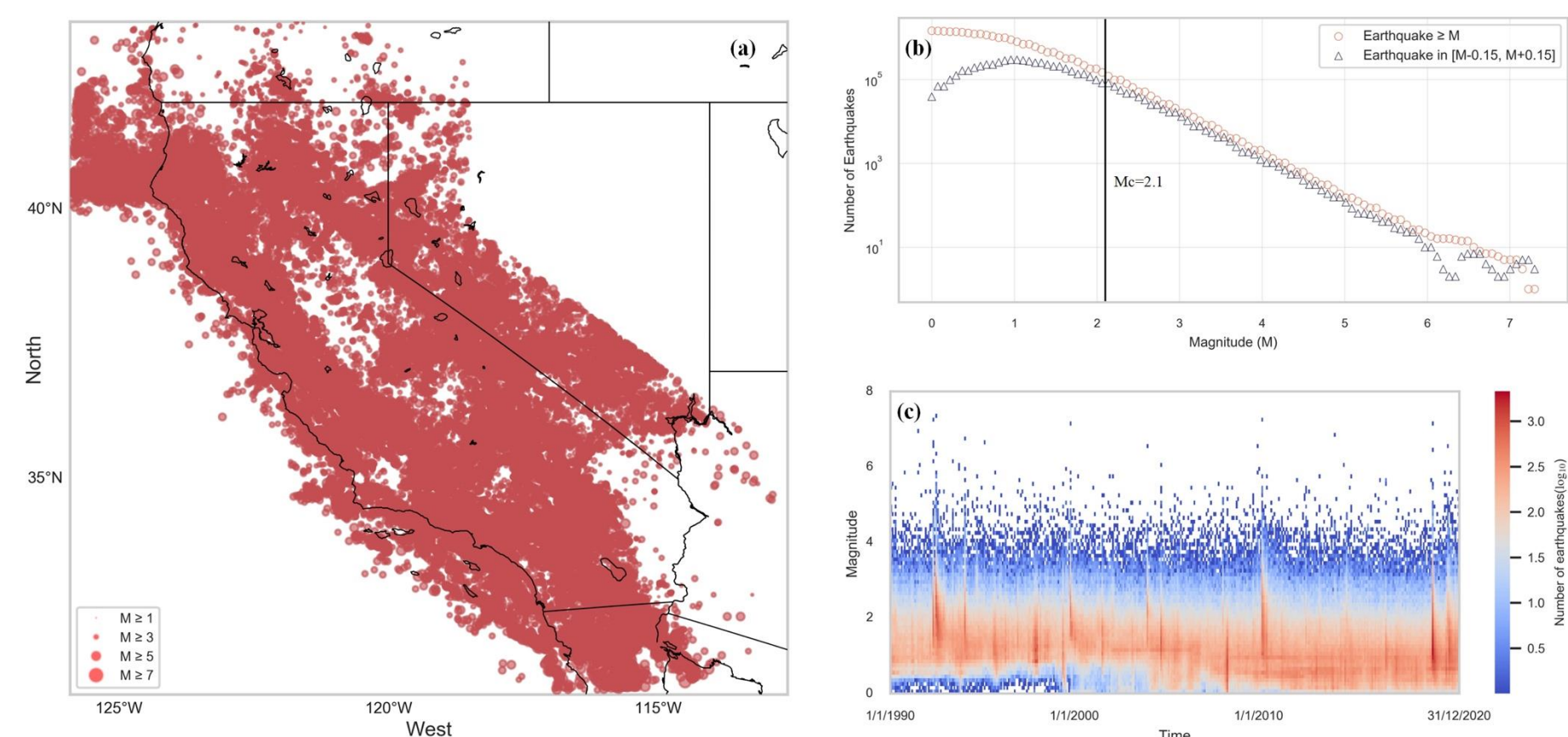
School of Science, China University of Geosciences (Beijing); The Institute of Statistical Mathematics

Introduction

Research on integrating statistical knowledge into deep learning models for earthquake forecasting has been limited. Traditional deep learning models require extensive parameter learning from scratch. This study proposes a spatio-temporal convolutional (STC) model that incorporates spatio-temporal decay prior knowledge derived from the Epidemic-Type Aftershock Sequence (ETAS) into a convolutional kernel. This allows the STC model to have the prototype to learn the pattern of mainshocks to trigger aftershocks at the beginning of training, with only four neurons to fine-tune it. In California, the STC and the ETAS model are conducted for forecasting next-day earthquakes with magnitudes of $M \geq 3$, $M \geq 4$, and $M \geq 5$. Both performances were assessed using the receiver operating characteristic (ROC) curve, the precision-recall (PR) curve, and the parimutuel gambling score (PGS).

Study area and datasets

The earthquake catalog ($M \geq 0$, $40 \text{ km} \geq \text{depth} > 0$) was obtained from the Advanced National Seismic System (ANSS) in California between January 1, 1990 and December 31, 2020. The spatial extent of this study area is illustrated in Fig. 1(a). The global empirical frequency-magnitude distribution of the earthquake catalog is shown in Fig. 1(b), with the M_c estimated to be 2.1, as indicated by the black solid line. Fig. 1(c). Nandan et al. (2017) indicated that M_c is spatially variable and established that $M_c = 3$ is almost applicable to the whole of California. For parameter calibration of the ETAS model, only earthquakes ($M \geq 3$) were considered. However, the STC model was trained, validated, and tested using a variable magnitude threshold ranging from 0 to 3, inclusive.



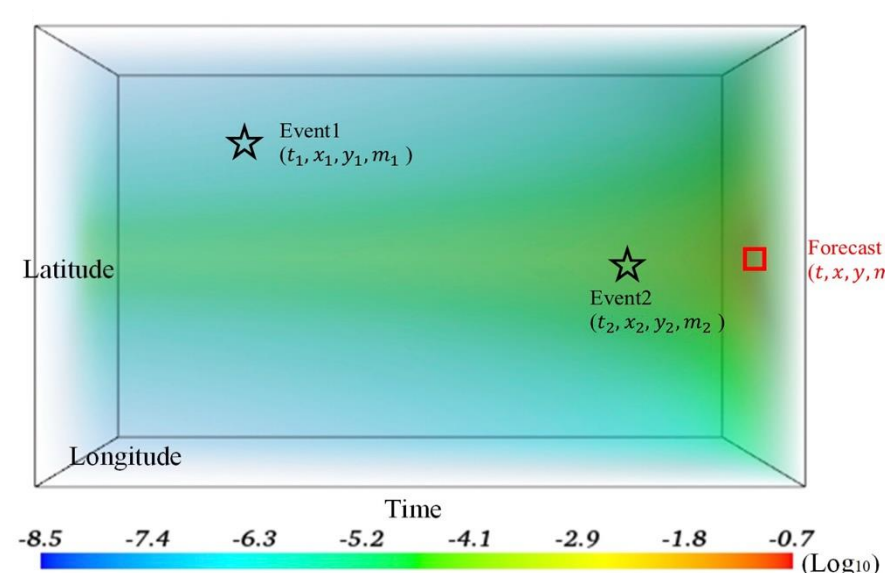
Construction of the STC model inspired by ETAS

We illustrate how the STC model is constructed inspired by the ETAS model by comparing and analyzing the details of the STC model with the ETAS model:

- For the background earthquakes, in the ETAS model, $\mu(x, y)$ represents the background intensity in space continuous. While in the STC model, the 2D matrix $\mu[x, y]$ represents the expected number of background earthquakes in space discrete.
- For the expectations of the number of aftershocks, in the ETAS model, as shown in Eq. (13), $k(m_i)$ indicates the expected number of earthquakes with magnitude m_i to produce aftershocks, where M_0 is the minimum magnitude that can generate aftershocks, K and a denote the parameters to be fitted. Correspondingly, in the STC model, as shown in Eq. (14), an input data matrix $I[i, j, k]$ represents the capability of aftershocks produced by earthquakes in that bin. In this process, the ETAS and STC model, M_0 is equal to M_c and 0, respectively.
- For the probability distributions of the time and location of the appearance of the aftershocks, denoted by a continuous time-space kernel function $(t - t_i, x - x_i, y - y_i, m_i)$ in the ETAS model, as shown in Eq. (15). The parameters c and ω control the temporal decay, while d, γ and ρ control the spatial decay. In the STC model, as shown in Eq. (16), turning time-space continuous functions $g(t - t_i, x - x_i, y - y_i, m_i)$ into the discrete 3D convolution kernels $C[t - i, x - j, y - k]$. Each value in the convolution kernel is calculated from the gridded spatio-temporal distance. c, p and d, q are neurons with an initial value equal to 1, which are used to learn the pattern of aftershocks decaying over time and space after the earthquake occurs, respectively.

To summarize the above, the ETAS model is based on the Hawkes process or self-exciting process, while the latter is known as the convolutional neural network. More detail can be found in that Table:

	ETAS		STC	
Calculation way	Hawkes point process	(1)	Discrete convolutional operation	(2)
Definition	$\lambda(t, x, y, m) \mathcal{H}_t = E(N(dt dx dy dm) \mathcal{H}_t)$	(3)	$\Lambda[t, x, y, m] \mathcal{H}_t = E(N[t, x, y, m] \mathcal{H}_t)$	(4)
Marginal intensity	$\lambda(t, x, y, m) \mathcal{H}_t = S(m) \lambda(t, x, y) \mathcal{H}_t$	(5)	$\Lambda[t, x, y, m] \mathcal{H}_t = J(m) \Lambda[t, x, y] \mathcal{H}_t$	(6)
Input volume	$\mathcal{H}_t = \{(t_i, x_i, y_i, m_i) : t_i \leq t\}$	(7)	$\mathcal{H}_t = \{(t_i, x_i, y_i, m_i) : t_i, x_i, y_i, m_i \in \Omega\}$	(8)
Conditional intensity function or its integral	$\lambda(t, x, y) = \mu(x, y) + \sum_{t_i < t} k(m_i) g(t - t_i, x - x_i, y - y_i, m_i)$	(9)	$\Lambda[t, x, y] = \beta \mu[x, y] + \alpha \left((I * C)[t, x, y] \right)$ $= \beta \mu[x, y] + \alpha \sum_{i,j,k} I[i, j, k] \cdot C[t - i, x - j, y - k]$	(10)
Background	$\mu(x, y)$	(11)	$\mu[x, y]$	(12)
Capacity to produce aftershocks	$k(m_i) = K e^{a(m_i - M_0)}$	(13)	$I[t, x, y] = \sum_{j,q \in [t,x,y]} e^{m_j}$	(14)
kernel function	$g(t - t_i, x - x_i, y - y_i, m_i) = \frac{1}{(t - t_i + c)^{1+\omega} ((x - x_i)^2 + (y - y_i)^2 + d e^{r(m_i - M_0)})^{1+\rho}}$	(15)	$C[t - i, x - j, y - k] = \frac{1}{(t - i + c)^p ((x - j)^2 + (y - k)^2 + d)^q}$	(16)
Optimization objectives	$\theta = \{\mu, K, a, c, \omega, d, \gamma, \rho\}$	(17)	$\theta = \{\alpha, \beta, d, q, c, p\}$	(18)
Optimization function	$l(\theta) = \sum_t \log(\lambda(t, x, y)) - \int_0^T \int_S \lambda(t, x, y) dx dy dt$	(19)	$loss = \text{mean}(\Lambda[t, x, y, m] - N[t, x, y, m])^2$	(20)
Probability conversion	Catalog simulation	(21)	$P_{STC}[t, x, y, m] = 1 - e^{-\Lambda[t, x, y, m]}$	(22)



This diagram denotes the spatio-temporal convolution kernel applied in STC model. The bins to be predicted are indicated by the red squares, while the historical earthquakes are represented by the black stars. The triggered component of 'Event 1' or 'Event 2' for the predicted bin is determined by the spatial and temporal intervals ($|t_1 - t|$, $|x_1 - x|$, $|y_1 - y|$) or ($|t_2 - t|$, $|x_2 - x|$, $|y_2 - y|$) and the magnitude m_1 or m_2 , respectively. Historical earthquakes closer in time and space to the forecasted bin have a greater triggered effect with the same magnitude.

Result

Fig. 2 (a) presents the spatial distributions of positive samples, and the output probabilities of both ETAS and STC models are shown in Fig. 2 (b), and (c), respectively. The output probability of STC is composed of radial triggered component (Fig. 2 (d)) with scattered background component (Fig. 2 (e)). The spatial distribution of positive samples in the training dataset is shown in Fig. 2 (f).

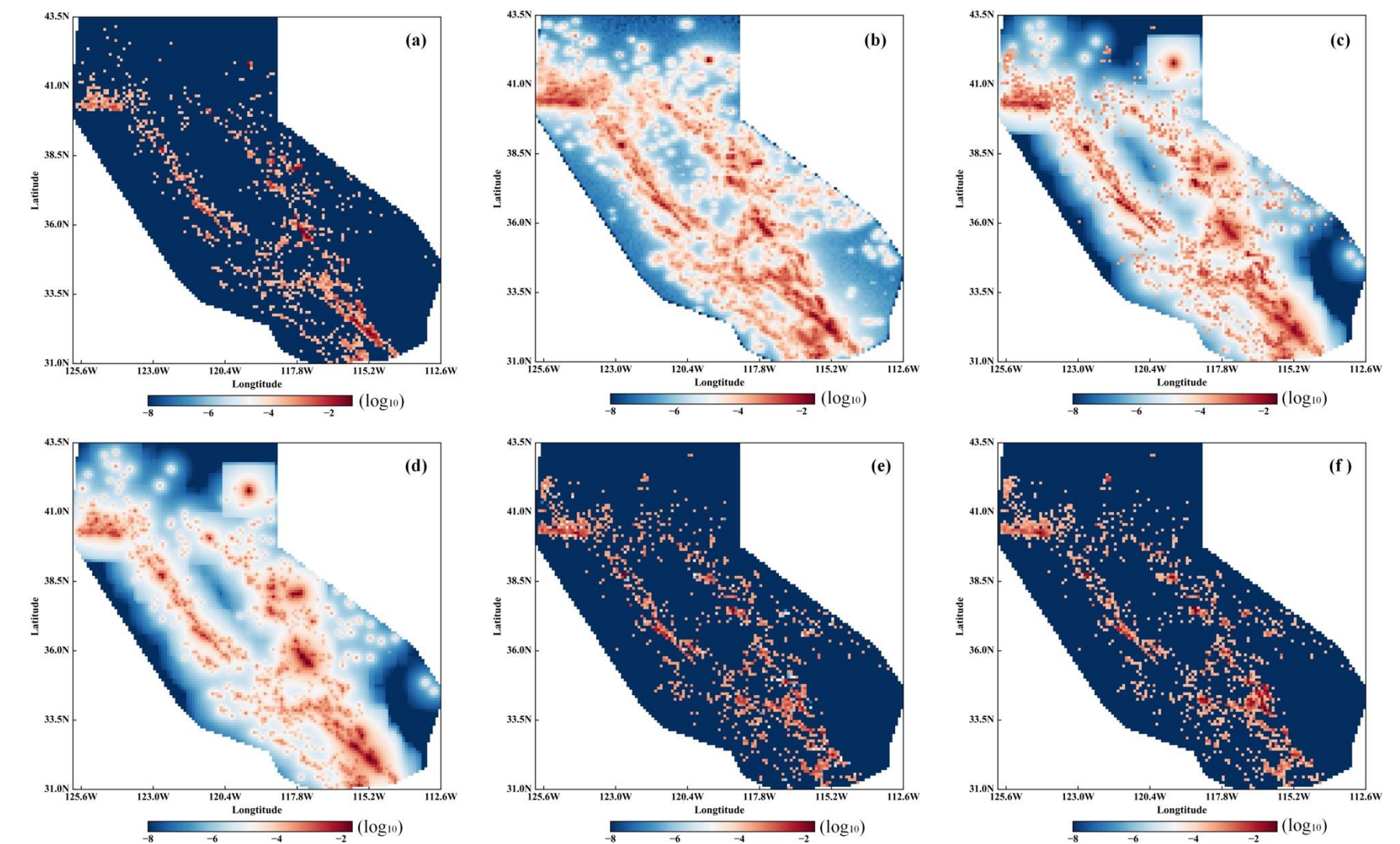
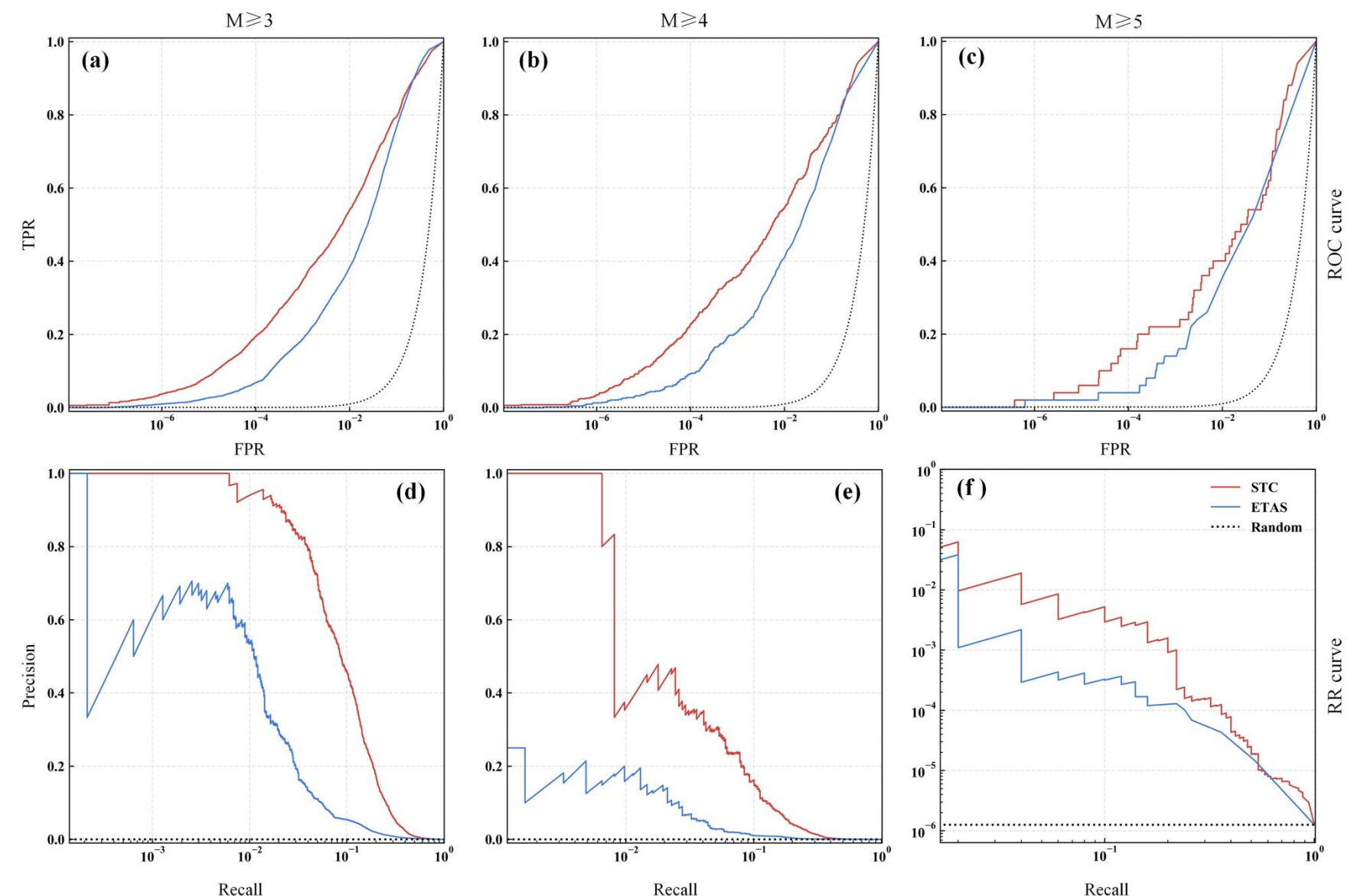
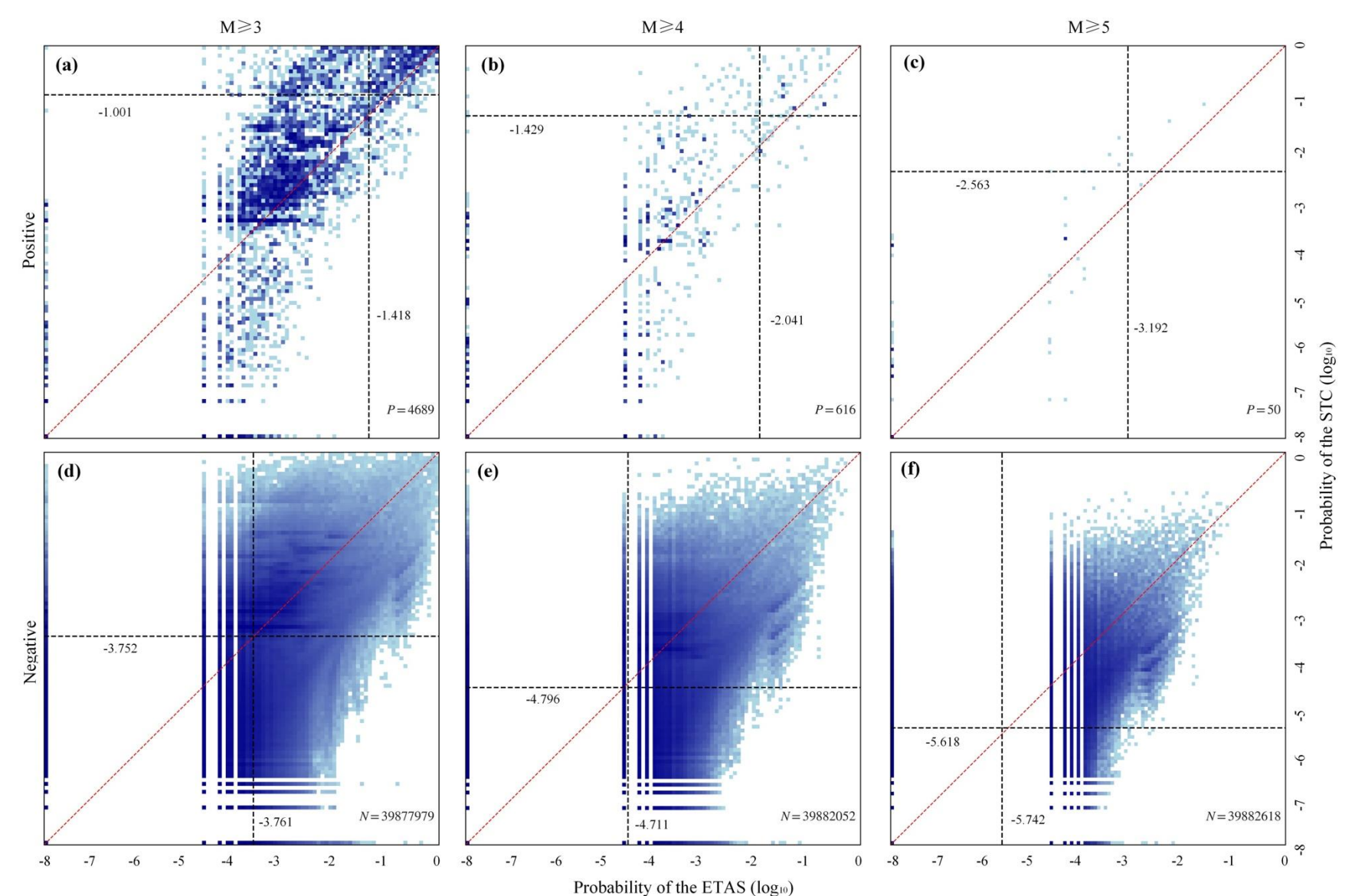


Fig. 3 presents ROC and PR curve comparisons between STC and ETAS under varying magnitude thresholds. STC exhibits markedly superior early-stage performance. For $M \geq 3$ (Fig. 3 (a)), it achieves lower FPR at equivalent TPR, outperforming ETAS up to a TPR of 0.9. Similar trends are observed for $M \geq 4$ and $M \geq 5$ (Figs. 3(b)–(c)). In PR analysis (Figs. 3(d)–(f)), STC attains $>90\%$ precision for the top 1% of events and maintains substantially higher AUC(PR) across all thresholds, especially at early recall stages. While precision declines with increasing recall due to negative sample dominance, STC consistently surpasses ETAS, particularly under stricter magnitude thresholds where prediction becomes more challenging.



We also assessed the probability forecasting capabilities of both models through the PGS. Fig. 4 (a), (b), and (c) demonstrate that the STC model assigns a higher average probability to positive samples compared to ETAS under the three target magnitudes, with PGS is 1154.848, 143.096, and 24.085, respectively. Conversely, for negative samples, both models yield similar average probabilities, as indicated in Fig. 4 (d), (e), and (f), with PGS is -44.558, 65.378, and -12.062, respectively.



Discussion & Conclusions

By integrating prior knowledge of ETAS model that aftershocks decay in time and space after the main shock into convolutional neural networks, the STC model was proposed to improve the performance and efficiency of earthquake forecasting. In contrast to complete deep learning approaches, the STC model benefits from the incorporation of prior knowledge, which not only enhances interpretability but also mitigates the risks of underfitting and overfitting. In addition, this approach inevitably imposes constraints on the flexibility of the neural network. The rigid integration of domain-specific assumptions may limit the model's capacity to learn alternative or unforeseen patterns inherent in the data. Future work may consider hybrid frameworks that strike a balance between incorporating prior knowledge and maintaining model adaptability, allowing neural networks to benefit from expert-driven guidance while retaining the capacity for data-driven discovery.